FRTN30 Assignment 2

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1 a)

We computed the shortest path with the python function *shortest path* from networkx, which uses Dijkstra's algorithm. Shortest path from node 1 to node 17 is:

$$1 \to 2 \to 3 \to 9 \to 13 \to 17 \tag{1}$$

Total travel time along this path is:

$$\sum_{e \in \text{path}} l_e = 0.532996 \tag{2}$$

2 b)

We computed the max-flow with respect to the capacities by using the *maximum flow* function from networkx. The theory builds upon Ford-Fulkerson method although that specific function uses the preflow-push algorithm. From the lecture notes Chapter 3.3 example 3.5 we know that Ford-Fulkerson uses the residual graph G_f where every edge (u, v) has a residual capacity $c_f(u, v)$ which is the remaining possible flow we could push through it. Starting from an initial feasible $o - d$ path and flow we define an augmenting path P from the origin o to the destination d in the residual graph such that:

$$c_f(u, v) > 0 \quad \forall (u, v) \in P.$$

For these augmenting paths P, we can increase flow by the lowest residual capacity value:

$$\Delta f = \min_{(u,v) \in P} c_f(u, v).$$

By keep doing iterations where we increase the flow along such paths (and update the residual graph with reverse edges so that we're able to undo earlier assigned flows) we eventually reach a point where no augmenting path is left and we have reached termination where the resulting flow is maximal. The maximum flow from node 1 to node 17 was computed to be:

$$\text{Max flow} = 22448$$

3 c)

Given the flow vector we computed the external in- or outflow by multiplying node-link incidence matrix with the flow vector.

$$\mu = Bf$$

which gave

$$\mu = \begin{bmatrix} 16806 \\ 8570 \\ 19448 \\ 4957 \\ -746 \\ 4768 \\ 413 \\ -2 \\ -5671 \\ 1169 \\ -5 \\ -7131 \\ -380 \\ -7412 \\ -7810 \\ -3430 \\ -23544 \end{bmatrix}$$

This works because the external flow will be the difference between the outgoing and ingoing flow, and since the **B** matrix keeps tracks of all out- and ingoing edges with distinction ± 1 all we have to do is multiply it with the flow along each edge.

4 d)

The social optimum f^* is the flow when the total travel time in the network is minimized. To find the total travel time we use:

$$\sum_{e \in \varepsilon} f_e d(f_e) \tag{3}$$

Where f_e is the flow of cars for a certain edge (road) and $d_e(f_e)$ is the delay function for that edge. The product $f_e d_e(f_e)$ gives us the total congestion for that edge and by summating all edges we get the total congestion, or the total time spent by all drivers on the network. The given delay-function is:

$$\sum_{e \in \epsilon} \left(\frac{l_e C_e}{1 - \frac{f_e}{C_e}} - l_e C_e\right) \tag{4}$$

Where l_e is the free-flow value, C_e the capacity. By minimizing this non linear convex problem we get the social optimum flow over each edge. To solve this, we need the constraint $Bf = v$, which basically is the conservation of flow at every node, incoming-outgoing flow = external flow. Furthermore, the flow must be greater than zero and not exceed the capacity $0 \leq f_e \leq C_e$. In the exercise, all external flow was set to zero except for node 1 and node 17. The value at these nodes

was the same as calculated above. When solving equation 4 above, we get $f^* = 25943.63$, which can be interpreted as the total time spent by all drivers in the network. The flow vector (Table 1), gives us the flow on a specific edges, required to fulfill the the social optimum.

Link	flow [vec/h]
1	6642.39
2	6059.08
3	3132.22
4	3132.21
5	10163.61
6	4638.53
7	3006.26
8	2542.68
9	3131.60
10	583.31
11	0.10
12	2926.76
13	0.01
14	3132.21
15	5525.07
16	2854.19
17	4886.47
18	2215.58
19	463.69
20	2337.85
21	3317.86
22	5655.72
23	2373.14
24	0.01
25	6414.17
26	5505.35
27	4886.48
28	4886.48

Table 1: The flow over each link corresponding to the social optimum

5 e)

When analyzing the Wardrop equilibrium, we examine the selfish-behavior of drivers, i.e., if each driver chooses the route that minimizes their travel time, without considering the effect of their decision on the overall network. To do so, we use the cost function:

$$\sum_{e \in \epsilon} \int_0^{f_e} d_e(s) ds \tag{5}$$

For the given delay function above, the integral becomes: $-l_e C_e \log(1 - \frac{f_e}{C_e})$. When minimizing equation 5, using the same constraints as above, we get the solution: 15729.61. This value and the social optimum are not directly comparable, since they are subject to different functions, as discussed above. However, we may expect a higher total travel time in the network for selfish behavior. When printing the flow vector (Table 2) for the Wardop equilibrium, we can see higher flows on certain links, which may result in higher congestion and increasing travel time.

Index	flow [vec/h]
1	6715.63
2	6715.63
3	2367.39
4	2367.39
5	10090.37
6	4645.41
7	2803.87
8	2283.63
9	3418.46
10	0.00
11	176.86
12	4171.38
13	0.00
14	2367.39
15	5444.96
16	2353.17
17	4933.33
18	1841.54
19	697.10
20	3036.55
21	3050.27
22	6086.82
23	2586.55
24	0.00
25	6918.73
26	4953.94
27	4933.33
28	4933.33

Table 2: The flow over each link corresponding to the Wardrop equilibrium

6 f)

If we introduce tolls on each link, the delay functions becomes: $d_e(f_e) + f_e^* d^{'}(f_e^*)$ where f^* is the social optimum. The integral in the Wardrop equilibrium, given in equation 5, now becomes: $-l_e C_e \log(1 - \frac{f_e}{C_e}) + \frac{f_e l_e f^*}{(1 - \frac{f_e^*}{C_e})^2}$ We introduce this toll since we want to capture the marginal cost in the system. When the driver act selfish, choose a route that minimize their travel time, they impose

congestion for others on that link. The differentiated the delay function:

$$\frac{d}{df_e}(f_e d_e(f_e)) = d_e(f_e) + f_e d^{'}(f_e) \tag{6}$$

Here, the last term captures the additional delay imposed on all other users when the flow increase. By adding this toll, we can make every driver act in a way that satisfies the social optimum. To reach the flow corresponding to the social optimum we use the f_e^* in the last term in 6. If we compute the Wardrop equilibrium in the same way as described above, with the toll included, the flow vector becomes:

Index	flow [vec/h]
1	6641.66
2	6058.62
3	3132.60
4	3132.60
5	10164.34
6	4637.53
7	3006.45
8	2542.45
9	3131.52
10	583.04
11	0.00
12	2926.02
13	0.00
14	3132.60
15	5526.81
16	2854.65
17	4886.29
18	2214.12
19	463.99
20	2336.96
21	3318.64
22	5655.60
23	2372.94
24	0.00
25	6414.17
26	5505.54
27	4886.29
28	4886.29

Table 3: The flow over each link corresponding to the Wardrop equilibrium with tolls

As can be seen in Table 3, the Wardop equilibrium with tolls become the same as the social optimum flow as in Table 1.

7 g)

We want to calculate and minimize the total additional delay compared to the travel time/delay at free-flow, i.e. $f_e l_e$. So we use the new cost function

$$c_e(f_e) = f_e \left(d_e(f_e) - l_e \right),$$

with the same delay function $d_e(f_e)$ as before. With this new cost function we then solved the problem the same way as in exercise **d)**, which gave the following results for the system optimum:

$$f^* = \begin{bmatrix} 6653.26,\ 5774.79,\ 3419.67,\ 3419.66,\ 10152.74,\ 4642.69,\ 3105.79,\ 2662.10, \\ 3009.10,\ 878.47,\ 0.01,\ 2355.12,\ 0.01,\ 3419.66,\ 5510.04,\ 3043.63, \\ 4881.79,\ 2415.38,\ 443.69,\ 2008.13,\ 3487.32,\ 5495.45,\ 2203.84,\ 0.00, \\ 6300.70,\ 5623.50,\ 4881.79,\ 4881.79 \end{bmatrix}$$

To construct tolls that coincide with the social optimum, we use the same reasoning as in exercise f). First we compute:

$$\frac{d}{df_e}(f_e d_e(f_e) - f_e l_e) = d_e(f_e) + f_e d^{'}(f_e) - l_e \tag{7}$$

If we substitute f_e in the second term by f_e^*, the Wardrop equilibrium will coincide with the social optimum, the reasoning is the same as above. The integral in the Wardrop equilibrium, given in equation 5, now becomes: $-l_e C_e \log(1 - \frac{f_e}{C_e}) + \frac{f_e l_e f^*}{(1 - \frac{f_e^*}{C_e})^2} - l_e f_e$ and the computed flow vector:

$$f^* = \begin{bmatrix} 6653.26,\ 5775.08,\ 3419.59,\ 3419.59,\ 10152.74,\ 4642.71,\ 3105.72,\ 2661.93, \\ 3009.11,\ 878.18,\ 0.00,\ 2355.49,\ 0.00,\ 3419.59,\ 5510.03,\ 3043.45, \\ 4881.75,\ 2415.17,\ 443.79,\ 2008.31,\ 3487.24,\ 5495.55,\ 2203.97,\ 0.00, \\ 6300.69,\ 5623.56,\ 4881.75,\ 4881.75 \end{bmatrix}$$

If we compare the two flow vectors above, we can see the effect of the tolls, and that they are calculated in such a manner that they affect the drivers in a way that corresponds to the social optimum.